

Swathi M
Software Developer

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SUMMARY

- Accomplished **Software Developer** with over 2 years of experience in designing, developing, and maintaining full-stack applications using **Java**, **Spring Boot**, **Hibernate**, and front-end technologies such as **React** and **Bootstrap**.
- Proficient in implementing **RESTful web services** with **Spring MVC** and **Spring WebFlux**, ensuring efficient routing, request handling, and seamless integration with front-end frameworks.
- Expertise in **Core Java** concepts, including **Collections Framework**, **Design Patterns**, **Multithreading**, **Concurrency (CompletableFuture, Executor Framework)**, and **Socket Programming**.
- Skilled in database management with **MySQL**, **PostgreSQL**, and **Oracle**, leveraging **SQL**, **JPQL**, **HQL**, and **Criteria API** for efficient backend-to-database mapping and query optimization.
- Experienced in containerization and orchestration tools like **Docker** and **Kubernetes**, enabling efficient deployment, scalability, and management of microservices in dynamic environments.
- Proficient in developing and consuming APIs using **Spring Boot**, **Spring Data JPA**, and **Spring Security**, ensuring robust authentication and authorization mechanisms for secure applications.
- Experienced in developing **Node.js** backend services, integrating third-party APIs, and leveraging **WebSockets** for real-time data streaming while building **RESTful APIs** with **Express.js** for Scalability and seamless integration.
- Hands-on experience with cloud platforms, including **AWS (EC2, S3, RDS, CloudWatch)** and **Google Cloud Platform (Compute Engine, Cloud SQL, Cloud Storage, Cloud Monitoring)**, for designing, deploying, and monitoring scalable cloud-based solutions.
- Skilled in automated testing frameworks such as **JUnit**, **Mockito**, **Selenium**, and **Cucumber**, ensuring application reliability through comprehensive unit, integration, and end-to-end testing.
- Designed and optimized complex queries in **Snowflake** to enhance performance and reduced execution times across large datasets, while implementing query profiling and performance tuning to identify and resolve bottlenecks in data processing.
- Dedicated to writing clean, maintainable, and scalable code by adhering to **SOLID principles** and industry best practices, promoting long-term maintainability and high-quality solutions.
- Adept at **Agile methodologies**, fostering iterative development, continuous delivery, and effective collaboration with cross-functional teams to achieve project success.

SKILLS & INTERESTS

- Programming Languages:** Java, C/C++, Python, JavaScript, SQL.
- Web Technologies:** Spring Boot, Spring MVC, Spring Security, Hibernate, JSP, Servlets, ReactJS (with Redux), HTML5, CSS3, Bootstrap, RESTful APIs, Microservices, AJAX, JSON.
- Cloud Computing and Deployment Tools:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Docker, Kubernetes, Jenkins, Kafka.
- Database Technologies:** MySQL, PostgreSQL, MongoDB, Oracle Database, Snowflake.
- Development Tools:** IntelliJ IDEA, Eclipse, Visual Studio Code, MySQL Workbench.
- Project Management and Methodologies:** Agile (Scrum), Waterfall, SDLC.
- Testing:** JUnit, Mockito, Selenium, Cucumber, Postman, Unit Testing.
- Version Control and Build Tools:** Git, Maven, Gradle.

RELEVANT WORK EXPERIENCE

Java Developer
PNC FINANCIAL

Jun 2024 – Present

- Successfully migrated millions of records from a legacy system to a new firm-wide platform using Designed, developed, and maintained robust microservices architectures using **Java 11/17**, **Spring Boot**, and **Spring Cloud**, resulting in a 30% increase in application Scalability and Performance.
- Collaborated with data engineering teams to enhance query structures in **Snowflake**, ensuring adherence to best practices, and troubleshoot and resolved query timeout issues by adjusting warehouse sizes and optimizing resource utilization.
- Developed and exposed **RESTful APIs**, ensuring seamless integration with front-end applications and third-party services, utilizing **Swagger** and **Postman** for API documentation and testing.
- Developed and maintained **Node.js** backend services, building **RESTful APIs** with **Express.js** to ensure efficient performance, scalability, and seamless frontend-backend integration.

- Debugged and resolved performance issues in **jQuery** scripts, reducing response times by 20% and incorporated **jQuery** animations to create visually appealing transitions in user interfaces.
- Integrated authentication and authorization mechanisms using **Spring Security, OAuth 2.0, and JWT**, improving application security and data protection.
- Used **Node.js** with Express to create scalable microservices for enterprise applications and deployed **Node.js** applications to **AWS Lambda** for efficient and serverless operation.
- Optimized SQL and NoSQL database operations by utilizing **PostgreSQL, MySQL, and MongoDB**, reducing query execution times by 35% through proper indexing and query tuning.
- Automated CI/CD pipelines using **Jenkins**, ensuring smooth deployment cycles, used for implementing automated testing in CI/CD pipelines, increasing coverage.
- Automated the build and deployment process using **Jenkins, GitLab CI/CD, Docker, and Kubernetes**, reducing deployment time by 50% and improving development efficiency.
- Leveraged cloud platforms such as **AWS(EC2, S3, RDS, Lambda, API Gateway), Azure(Azure Functions, Cosmos DB) and Google Cloud Platform(GKE, Firestore)** to deploy and monitor cloud-native applications.
- Implemented real-time messaging and event-driven architecture using **Apache Kafka and RabbitMQ**, improving inter-service communication and reducing latency by 40%.
- Developed interactive and responsive UI components using **ReactJS, Redux, TypeScript, CSS3**, enhancing user engagement and system usability.
- Created triggers in **SQL Server** to handle real-time data updates and maintain data integrity and tuned query performance in **SQL Server**, reducing execution times by 30%.
- Ensured code quality and application stability by writing unit and integration tests with **JUnit, Mockito, TestNG, Selenium, and Cucumber**, achieving 95% test coverage.
- Performed container orchestration and cluster management using **Kubernetes (AKS, EKS, GKE)**, ensuring high availability and fault tolerance.
- Engaged in Agile methodologies, conducting daily stand-ups, sprint planning, and retrospective meetings using **JIRA** to track and document project progress.

Environment: Java 11/17, Spring Boot, Spring Cloud, Hibernate, JPA, PostgreSQL, MySQL, MongoDB, AWS (EC2, S3, RDS, Lambda, API Gateway), Azure, GCP, Kubernetes (AKS, EKS, GKE), Docker, Jenkins, GitLab CI/CD, Apache Kafka, RabbitMQ, ReactJS, Redux, TypeScript, Snowflake, CSS3, JUnit, Mockito, Selenium, Cucumber, Agile (Scrum), JIRA, jQuery, Node.js, SQL Server.

Java Developer
Amerispace Private Limited

Hyderabad,IND
Jan 2021 – Dec 2022

- Successfully migrated millions of records from a legacy system to a new firm-wide platform using **Spring Boot, Hibernate**, and **RESTful APIs**, improving API response times by 30% and reducing operational overhead by 25%.
- Utilized **jQuery** to develop dynamic, interactive web components, enhancing user engagement and experience and optimized DOM manipulation and event handling with **jQuery**, reducing page load times by 15%.
- Optimized data partitioning and clustering in **Snowflake** to improve query performance and data retrieval speeds, while resolving query performance issues by rewriting suboptimal SQL and fine-tuning join strategies.
- Streamlined backend processes and optimized database performance using **JPA, PostgreSQL**, and **MongoDB**, achieving 99.9% data accuracy and reducing query latency by 15%.
- Integrated third-party APIs and services within **Node.js** applications to enhance functionality. Leveraging **Websockets** for real-time data streaming and seamless inter-service communication.
- Improved deployment efficiency by 40% by utilizing Kubernetes for container orchestration and automating **CI/CD** pipelines with **AWS CodePipeline** and **CodeDeploy**, minimizing downtime and accelerating delivery timelines.
- Built **RESTful APIs** using **Node.js**, enabling efficient communication between front-end and back-end, optimized server-side performance by implementing asynchronous programming in **Node.js**.
- Enhanced user interfaces by developing new features and UI enhancements with **React** and **JavaScript**, leading to a 20% increase in end-user satisfaction and improved overall usability.
- Integrated **Apache Kafka** for asynchronous messaging, improving module communication efficiency by 35% and significantly enhancing message processing speed.
- Secured databases by configuring roles and permissions in **SQL Server**, and used **SQL Server** tools like Profile and Execution Plans for performance diagnostics.
- Ensured high code quality and reduced production defects by conducting comprehensive testing with frameworks like **JUnit** and **Mockito**, achieving robust and reliable application performance.
- Deployed containerized applications using **Docker** and managed infrastructure across **AWS** to ensure high scalability, reliability, and availability of services.
- Conducted performance and load testing with tools like **Postman** and **JMeter**, identifying bottlenecks and implementing optimizations to improve application responsiveness.

- Enhanced an existing application by making **API** changes and UI improvements, integrating **Spring Boot APIs** with a **ReactJS** front-end to improve usability and streamline data flow.
- Implemented a version comparison feature in the UI using **React** state management and **CSS** styling, allowing users to view changes with the previous version highlighted in red and the updated version in green, enhancing clarity and interaction.
- Optimized database operations with **JPA** and **PostgreSQL**, ensuring efficient storage and retrieval of version history while maintaining data consistency across the application.
- Achieved 98% test coverage and ensured high-quality, defect-free code by conducting unit and integration testing using **JUnit**, **Mockito**, and **React** Testing Library.

Environment: Java, Spring Boot, Hibernate, JPA, PostgreSQL, MongoDB, RESTful APIs, ReactJS, JavaScript, CSS, Apache Kafka, Kubernetes, Snowflake, Docker, AWS, JUnit, Mockito, React Testing Library, Postman, JMeter, jQuery, Node.js, SQL Server.

EDUCATION

● Roosevelt University	Chicago, Illinois
Master’s in Computer Science	CGPA : 3.69
● Malla Reddy Engineering College	Hyderabad,India
BTech in Computer Science and Engineering	CGPA : 7.89

PROJECTS

Spring Boot Web Application for Employee Role

- Developed a web application for managing employee roles and information, ensuring seamless CRUD operations and efficient data management.
- Designed and implemented a web application to manage employee roles and data using **Spring Boot** and **Hibernate** for efficient data persistence.
- Built **RESTful APIs** to handle CRUD operations on employee data, ensuring seamless interaction between front-end and back-end.
- Developed user-friendly interfaces using **HTML**, **CSS**, and **JavaScript**, enhancing usability and accessibility.
- Conducted unit testing to ensure module functionality and reliability, documenting test cases and results for clarity.
- Deployed the application locally for demonstrations and collaborated with peers to identify and resolve design and coding issues.
- Followed **SDLC best practices** throughout the project lifecycle, ensuring high-quality deliverables.

Technologies Used: Spring Boot, Hibernate, MySQL, HTML, CSS, JavaScript